

Computational Thinking with Python

Session 0 – Introduction

AGENDA

- 1 About
- 2 Fundamentals
- 3 PRIMM Framework
- 4 Getting Started

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MEET YOUR INSTRUCTOR

Nikolai Stein

- Senior Researcher (JMU Würzburg)
- Product Manager (SOLISTIQ)
- Visiting Professor (Aarhus University)
- Lecturer (Mannheim Business School)
- Lecturer (Gutenberg School of Business)

Background

- Bachelor (Business Management)
- Master (Information Systems)
- PhD (Business Analytics)

ABOUT THIS COURSE

- This course is designed for students with **little or no experience** of programming
- Broad introduction to Python programming and computational thinking
- We choose to focus on **breadth rather than depth**
- You will get an introduction to many topics:
 - Week 1 – Python fundamentals
 - Week 2 – Computational thinking, and web scraping
 - Week 3 – Data analysis and simulation

TELL US ABOUT YOU!

- What is your background?
- Do you have any programming experience?
- What do you study?
- ...

Let's get to know each other!



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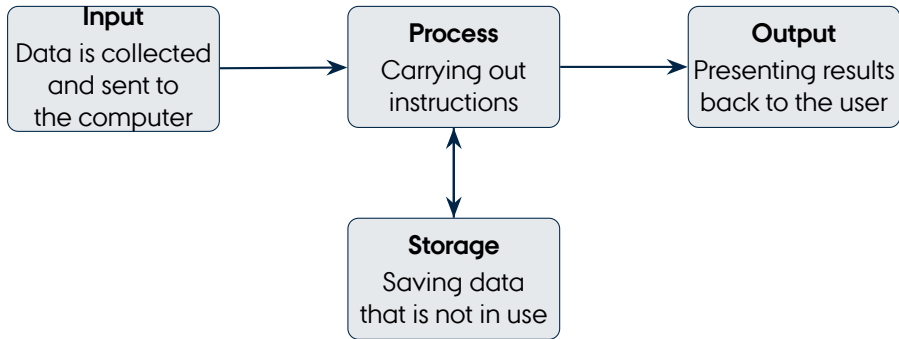
PROGRAMMING ENABLES YOU TO MAKE COMPUTERS SOLVE PROBLEMS

“Programming is how *you* get computers to solve problems.”

Two key concepts:

- **You:** Without the programmer (you), the computer is useless. It does what **you** tell it to do.
- **Solve problems:** Computers are tools. They exist to make tasks easier – they are not mysterious or magical.

COMPUTERS PROCESS DATA IN A SIMPLE CYCLE



AN ALGORITHM DESCRIBES THE LOGIC; A PROGRAM IMPLEMENTS IT

“An algorithm is a sequence of instructions for a task”

Algorithm $\neq$ Computer Program

- An algorithm describes the *logic* of a solution
- A computer program is the *implementation* in a specific language
- The same algorithm can be implemented in many programming languages

ALGORITHMS ARE EVERYWHERE IN DAILY LIFE

- **Making tea:** Boil water, add tea bag, wait, remove bag, add milk/sugar
- **Playing Tic-Tac-Toe:** Strategy for winning or drawing
- **Making a sandwich:** Layer ingredients in the right order

Video: <https://www.youtube.com/watch?v=FN2RM-CHkuI>

THE SAME ALGORITHM CAN BE WRITTEN IN DIFFERENT LANGUAGES

Algorithm (Pseudocode)

```
OUTPUT "Hello, World"
```

The same algorithm can be implemented in different languages with different syntax.

Python

```
print("Hello, World!")
```

JavaScript

```
console.log("Hello, World!");
```

Java

```
System.out.println("Hello,  
World!");
```

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PRIMM GUIDES LEARNING FROM READING CODE TO WRITING CODE

We use the **PRIMM framework** throughout this course:

- **Predict:** Look at working code. What do you think it will do?
- **Run:** Execute the code and test your prediction
- **Investigate:** What does each line mean? Trace through the code.
- **Modify:** Edit the program to make it do different things
- **Make:** Design a new program using the same concepts

PRIMM REDUCES COGNITIVE LOAD FOR BEGINNERS.

Traditional Approach

- Start with blank screen
- Write code from scratch
- High cognitive load
- Often frustrating for beginners

PRIMM Approach

- Start with working code
- Build understanding gradually
- Lower barrier to entry
- Builds confidence step by step

Key Takeaway: As you progress from Predict to Make, you take more **ownership** of the code.

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GITHUB CODESPACES LETS YOU CODE DIRECTLY IN YOUR BROWSER

- This will be a **hands-on class!**
- We implement the PRIMM framework using **GitHub Codespaces**
- GitHub Codespaces provides an instant IDE in your browser
- Create an account to access code templates and exercises

No installation required – just a web browser!

CREATE A GITHUB ACCOUNT AND ACCESS THE COURSE REPOSITORY

1. Create a **GitHub account** at <https://github.com>
2. Access the course repository (Link on Brightspace)
3. Open the first exercise in GitHub Codespaces
4. Start with Session 1: Python Basics!

Questions? Ask now or reach out via email!

Thank You!

Nikolai Stein

